BrainAI

ProSuite Industrial

INDUSTRIAL ENCRYPTION SUITE

PRODUCT DOCUMENTATION

BrainAI (Inhaber: Sascha Alexander Köhne)

BrainAI Inhaber: Sascha Alexander Köhne	PROSUITE INDUSTRIAL	🟢 PUBLIC CLASSIFICATION
Core Architecture & Development	**ProEDC CLI Reference**	Access Level: Public / Beta

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Revision History

Version	Date	Author	Status	Description
1.0.0	2026-01-01	S. Köhne	Stable	Initial Commercial Suite CLI
1.1.0	2026-02-26	S. Köhne	Stable	Hybrid Ring-0, Audit-Logging & Secure Wipe (PRO Edition)

1. Product Overview

The **ProEDC CLI Suite** is the primary administration tool for the commercial ProED series. Unlike the standard version, ProEDC integrates a complete license management system, advanced cryptographic utility functions (KeyGen, Hashing), and a tamper-proof audit trail.

Core Features:

- **Hybrid Acceleration (PRO Only):** Automatic detection and utilization of the `ProKM` kernel driver for lossless Ring-0 performance.
- **Audit-Proof Logging:** Optional logging of operations, including cryptographic input and output hashes for seamless traceability.
- **Secure Data Shredding (PRO Only):** Multiple overwriting of source files after successful processing to meet the highest data protection standards.
- **Auto-Keygen:** Automatic creation of secure 256-bit keys if parameters are missing during the encryption process.
- **Integrated Suite:** Combines file encryption, key generation, and ProHash verification in a single binary.

2. Technical Specifications

2.1 Command-Line Parameters (CLI)

The suite follows the syntax: `proedc [MODE] [OPTIONS]`.

Modes:

- `-e, --encrypt`: Encryption mode.
- `-d, --decrypt`: Decryption mode.
- `-g, --keygen`: Generation of cryptographic keys.
- `--hash`: Creation of ProHash values for files or strings.

Options:

- `-i, --input <path>`: Path to the input file.
- `-o, --output <path>`: Path to the output file.
- `-k, --key <path>`: Path to the key file.

- `-s, --string <text>`: Text for direct hashing (without a file).
- `-b, --bits <n>`: Key length in bits (Default: 256).
- `-L, --log`: **(PRO)** Enables detailed audit logging.
- `-D, --delete`: **(PRO)** Securely wipe the source file after success.

3. Global Error Management

ProEDC uses an expanded status model that accounts for license and hardware states.

Exit Code	Constant	Description	Criticality
0	PEDC_SUCCESS	Operation completed successfully.	None
1	PEDC_ERR_PARAM	Invalid parameters or missing arguments.	Medium
2	PEDC_ERR_AUTH	MAC check failed (file manipulated).	High
3	PEDC_ERR_FILE_IO	Error reading/writing files.	Medium
-7	PEDC_ERR_LIMIT	License limit reached (e.g., TRIAL expiration).	Fatal

4. Implementation Examples

4.1 Encryption with Auto-Keygen & Audit Log (PRO)

In this example, a file is encrypted. Since no key is specified, ProEDC automatically creates one and logs all hashes.

```
# Execution in the PRO Edition
./proedc -e -i confidential.dat -o confidential.enc -L

# Result in proedc_audit.log:
# [2026-02-26 14:30:00] OP: AUTO-KEY | Status: OK | IN: N/A | OUT: au
# [2026-02-26 14:30:00] OP: ENCRYPT  | Status: OK | IN: confidential.
```

4.2 Decryption followed by Shredding (PRO)

This command decrypts a file using the kernel bypass and irretrievably deletes the encrypted source file after success.

```
./proedc -d -i backup.enc -o backup.zip -k master.key -D
# [+] PRO Edition: Hardware Kernel Driver (ProKM) detected.
# [+] PRO Feature: Source file securely wiped (-D).
```

4.3 String Hashing for Configurations

Direct generation of a ProHash value for a text string:

```
./proedc --hash -s "Master-Admin-Password-2026"
# String Hash: 7f83b165...
```

5. Best Practices & Security Guidelines

- **Audit Compliance:** In automated environments, always use the `-L` flag to be able to prove the integrity of the data chain during security audits.
- **Data Hygiene:** In the PRO Edition, use the `-D` flag for temporary files to ensure that no fragments of sensitive data remain on physical storage media.
- **License Monitoring:** Regularly check exit code `-7` to react in time before TRIAL licenses or volume limits expire in production systems.
- **Kernel Privileges:** To use Ring-0 acceleration, the executable must have sufficient permissions to communicate with the `ProKM` device.

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CEO / System Architect	BrainAI Inhaber : Sascha Alexander Köhne	We don't need **BRUTEFORCE**, we know **PHYSICS**